Product Requirements Document

"FundLens" — Personal Mutual Fund Portfolio Analyzer

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1. Vision & Problem Statement

Retail MF investors in India hold portfolios scattered across platforms (Groww, Zerodha Coin, direct AMC, Dezerv, NSDL demat). No single tool answers the questions that actually matter to an investor:

- What is my *true* return (XIRR), fund-by-fund and overall?
- Am I beating a passive benchmark, or paying fees to underperform an index fund?
- Which funds should I hold, which should I exit — and what does exiting cost me in tax?
- Am I on track for my target corpus given my risk appetite?

Zerodha Coin's benchmark comparison and Kuvera's XIRR are partial answers locked inside single platforms. **FundLens** is a platform-agnostic, privacy-first analyzer that ingests the investor's complete history, computes honest performance metrics, benchmarks against indices, and produces expert-grade, AI-assisted hold/sell recommendations with tax awareness.

Primary user (v1): A single sophisticated retail investor (self). Multi-user is out of scope for v1 but the data model should not preclude it.

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2. Goals & Non-Goals

Goals

1. Single source of truth for all MF holdings and full transaction history across platforms.
2. Accurate return metrics: XIRR, CAGR, absolute return — at transaction, fund, category, platform, and portfolio level.
3. Benchmark comparison (BSE 100 TRI, Nifty 50 TRI, category-appropriate indices) the way Zerodha Coin does, but across the *whole* portfolio.
4. Actionable per-fund verdicts: Hold / Sell / Switch, with tax-impact-aware reasoning.
5. Risk-profile-driven asset allocation targets and drift alerts.
6. Goal planning: target corpus, required SIP, probability of achieving it.
7. Yearly (and on-demand) reports, including capital gains for ITR filing.
8. AI-powered qualitative analysis layered on top of deterministic metrics.

Non-Goals (v1)

- Executing transactions (buy/sell/switch). Analysis only.
- Stocks, bonds, FDs, EPF/NPS as first-class assets (may appear as "external corpus" input

for goal math).

- Multi-tenant SaaS, mobile app. v1 is a local/self-hosted tool.
- Being a SEBI-registered advisory product (see §11, Compliance).

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3. Personas & Risk Profiles

Persona	Risk appetite	Equity:Debt target	Behavior
Conservative	Low	30:70	Capital preservation, avoids mid/small cap
Balanced	Medium	60:40	Index-heavy core, modest satellite
Aggressive	High	80:20	Mid/small-cap tilt, long horizon

Persona	Risk appetite	Equity:Debt target	Behavior
Custom	User-defined	Any	Power user sets own bands per bucket

The risk profile is a first-class object: it drives allocation targets, drift alerts, recommendation tone, and alternative suggestions. The user selects or customizes it during onboarding via a short questionnaire (horizon, drawdown tolerance, income stability, dependents).

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4. Functional Requirements

FR-1: Data Ingestion

FR-1.1 — CAS (Consolidated Account Statement) parsing [P0]

- Ingest MFCentral / CAMS+KFintech detailed CAS PDFs (password-protected) using `casparser` or equivalent.
- Extract: folio, AMC, scheme (with AMFI/ISIN code), transaction type (purchase, SIP,

redemption, switch in/out, dividend payout/reinvest, segregation), units, NAV, amount, date, stamp duty, STT.

- Handle password-protected PDFs (batch decryption supported).
- **The detailed CAS with full history is the canonical source of truth.** It covers all platforms (Groww, Coin, Dezerv, direct) because it is registrar-level data.

FR-1.2 — Incremental updates [P0]

- Re-uploading a newer CAS must merge idempotently: dedupe transactions by (folio, scheme, date, type, amount, units) natural key; never double-count.
- Detect and surface gaps (e.g., CAS date range doesn't cover a known folio's start).

FR-1.3 — Platform scraping/import (Groww, MFCentral web) [P2, de-scoped with rationale]

- *PM note:* Scraping Groww/MFCentral violates their ToS, breaks on every UI change, and is redundant — the detailed CAS already contains 100% of registrar-recorded transactions. Recommendation: **do not build scrapers.** Instead support (a) monthly CAS auto-fetch

reminder, (b) manual CSV import as an escape hatch for exotic cases (e.g., international feeder funds not in CAS).

FR-1.4 — Market data [P0]

- Daily NAVs: AMFI NAVAll feed (free, official) or api.mfapi.in; store full NAV history per held scheme.
- Benchmark index data: BSE 100 TRI, Nifty 50 TRI, Nifty Midcap 150 TRI, Nifty Smallcap 250 TRI, category-appropriate debt indices. **Must use Total Return Index (TRI), not price index** — comparing fund NAVs (which include dividends) against a price index flatters the fund. This is the same discipline Zerodha Coin follows.
- Scheme metadata: category (SEBI classification), expense ratio, AUM, fund manager, benchmark declared by the AMC.

FR-1.5 — Manual adjustments [P1]

- Allow marking a scheme as "ignore" (e.g., closed folios) and tagging holdings to goals. Every folio is tagged to a PAN at ingestion (auto from CAS, manually correctable). External non-MF corpus is out of scope per §12.

FR-2: Portfolio & History

FR-2.1 [P0] Reconstruct complete investment history: every transaction, running units, cost basis per lot (FIFO), current value from latest NAV.
FR-2.2 [P0] Views: by PAN (primary partition, with household roll-up), by fund, by AMC, by category (large/mid/small/flexi/debt/gold/international), by platform/folio, by goal, by tax status (LTCG-eligible vs STCG).
FR-2.3 [P1] SIP detection: infer active SIPs (recurring same-amount purchases), show SIP calendar and monthly outflow.
FR-2.4 [P1] Corporate-action correctness: switch-out/switch-in pairs, dividend reinvestment, segregated portfolios (side-pocketing) must not corrupt XIRR.

FR-3: Performance Metrics

FR-3.1 — XIRR [P0]

- Per fund, per folio, per category, per goal, per platform, and portfolio-wide.
- Cash flows = all purchases (negative), redemptions/dividend payouts (positive), current value as terminal flow.

- Edge cases: fully-exited funds (realized XIRR), funds < 1 year old (show absolute return, label XIRR as annualized-and-noisy), zero/negative-flow degenerate cases.
  FR-3.2 [P0] Absolute return, CAGR (point-to-point), invested vs current value, realized vs unrealized P&L.
  FR-3.3 [P1] Rolling returns (1/3/5-year) for held funds vs benchmark — better than point-to-point for judging consistency.
  FR-3.4 [P2] Risk-adjusted metrics: Sharpe, Sortino, max drawdown, standard deviation vs category peers.

FR-4: Benchmark Comparison ("Zerodha-style")

FR-4.1 [P0] For every fund: "What if the same cash flows had gone into the benchmark TRI?" — compute benchmark XIRR with identical dates/amounts and show fund XIRR vs benchmark XIRR side by side, with alpha (±).
FR-4.2 [P0] Portfolio-level: same simulation against a single user-chosen reference (**default: Nifty 50 TRI**, per §12), computed per PAN and for the household roll-up view.
FR-4.3 [P1] Category-correct benchmarks by

default (midcap fund → Midcap 150 TRI), with override.
FR-4.4 [P1] "Index fund counterfactual": for each active fund, compare against the cheapest index fund tracking its benchmark (fee-adjusted), not just the raw index.

FR-5: AI-Powered Expert Analysis

FR-5.1 [P0] An LLM analysis layer (Claude API) that receives *structured, pre-computed* metrics — never raw PDFs — and produces:

- Portfolio health narrative (concentration, overlap, style drift, fee drag).
- Per-fund qualitative commentary (manager changes, category headwinds, consistency).
  FR-5.2 [P0] Determinism boundary: **all numbers come from the compute engine; the LLM interprets, it never calculates.** LLM output must cite which metric drove each statement.
  FR-5.3 [P1] Fund overlap analysis: pairwise portfolio-holding overlap % across equity funds (data source: monthly AMC portfolio disclosures) to expose structural duplication.
  FR-5.4 [P2] News/context enrichment via web search (fund manager exits, AMC events) with sources shown.

FR-6: Recommendations (Hold / Sell / Switch)

FR-6.1 [P0] Per-fund verdict with a scored rubric (transparent, tunable weights):

- 3-yr & 5-yr alpha vs category benchmark (rolling)
- Consistency (rolling-return win rate vs benchmark)
- Expense ratio vs category median; direct-vs-regular check
- Overlap/redundancy with rest of portfolio
- Category fit vs user's risk profile & target allocation
  FR-6.2 [P0] Every SELL must include **tax and friction impact**: STCG vs LTCG per lot, LTCG exemption headroom used, exit load, and a "net-of-tax breakeven" statement ("switching costs ₹X now; new fund must outperform by Y%/yr to justify").
  FR-6.3 [P1] Action plan generator: ordered, tax-optimized exit/switch sequence (e.g., harvest within ₹1.25L LTCG exemption each FY, exit STCG lots after they turn LTCG).
  FR-6.4 [P1] Verdicts are advisory-grade explanations, not one-word labels: each carries

reasoning, confidence, and "what would change this verdict."

FR-7: Risk Appetite & Allocation

FR-7.1 [P0] Compare actual allocation (equity/debt/gold/international; large/mid/small within equity) against the risk profile's target bands; show drift %.
FR-7.2 [P1] Drift alerts and rebalancing suggestions (prefer directing *new* SIP money over selling, to avoid tax).
FR-7.3 [P2] Age/horizon-based glide path suggestion (equity % steps down as goal date approaches).

FR-8: Target Corpus & Goal Planning

FR-8.1 [P0] Define goals: name, target amount (with inflation assumption), target date, tagged MF holdings (can span PANs; corpus math is MF-only per §12).
FR-8.2 [P0] For each goal: current corpus, projected corpus at expected return (risk-profile-based), required monthly SIP to close the gap, and sensitivity table (return ±2%).
FR-8.3 [P2] Monte Carlo simulation: probability of

reaching target given historical return/volatility of the actual allocation.

FR-9: Alternatives Engine

FR-9.1 [P1] For every SELL/SWITCH verdict, suggest 2–3 alternatives from the *same category*, ranked by: rolling-return consistency, expense ratio, AUM sanity, overlap with existing holdings. Default bias toward low-cost index funds where the category is index-replicable.
FR-9.2 [P2] "Simplification mode": propose a minimal 4–6 fund portfolio equivalent (core index + satellite) and quantify the fee savings per year in ₹.

FR-10: Reports

FR-10.1 [P0] Yearly report (FY-aligned, Apr–Mar): opening/closing value, net invested, XIRR for the year and since inception, benchmark comparison, best/worst funds, allocation drift, fees paid (estimated from expense ratios), realized gains.
FR-10.2 [P0] Capital gains report for ITR: scheme-wise STCG/LTCG with grandfathering handled, exportable CSV/PDF matching ITR schedule-112A shape.
FR-10.3 [P1] On-demand snapshot report

(PDF/HTML) and a monthly one-pager.
FR-10.4 [P2] Email/notification digest.

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5. Additional Requirements (PM additions — beyond the original brief)

These are the requirements I'd insist on as a product owner; several are the difference between a toy and a trustworthy tool:

1. **Tax engine as a first-class module [P0].** Indian MF taxation (equity vs debt vs hybrid rules, holding periods, LTCG exemption, grandfathering, post-2023 debt rules) changes recommendations materially. Without it, "sell" advice is actively harmful.
2. **Overlap & duplication detection [P1].** (FR-5.3) Your own portfolio review found index/gold duplication — this is a systemic retail problem worth automating.
3. **Direct-vs-Regular audit [P0].** Flag every regular-plan holding, quantify trailing commission drag in ₹/year, since it's the single most common fixable leak.

4. **Data quality dashboard [P1].** Show ingestion coverage: folios found, date ranges, unmatched schemes, XIRR computations that failed — trust requires visible provenance.
5. **Privacy-first architecture [P0].** All financial data stays local (SQLite, encrypted at rest). Only anonymized, aggregated metrics leave the machine (to the LLM API). No account credentials ever stored; CAS PDFs are pull-based by the user.
6. **Explainability everywhere [P0].** Every number drillable to its underlying transactions; every verdict drillable to its rubric scores.
7. **Assumption registry [P1].** Inflation %, expected returns, exemption limits are editable named assumptions, versioned per report, so two reports are comparable.
8. **Idempotent, replayable pipeline [P1].** Raw CAS files are immutable inputs; the entire derived state can be rebuilt from raw files + NAV history (event-sourcing style). This makes bugs fixable retroactively.
9. **Watchlist [P2].** Track funds you don't own (candidates from the alternatives engine) with the same metrics.

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6. Non-Functional Requirements

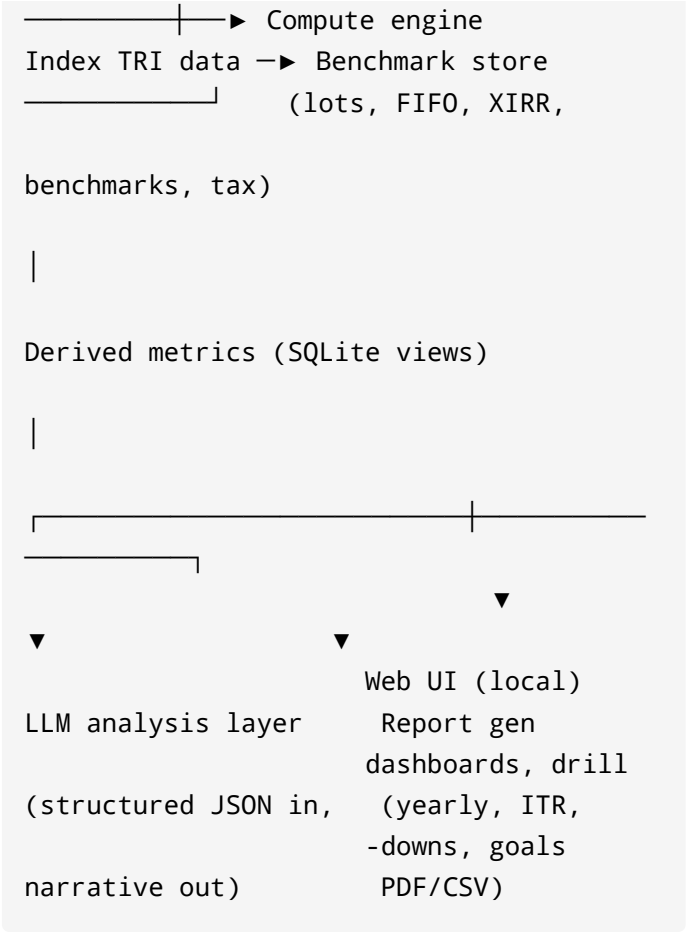
- **Accuracy:** XIRR within 0.01% of Excel's XIRR on identical cash flows (golden test suite). Capital-gains report reconciles with CAMS/KFintech capital gains statement within rounding.
- **Performance:** Full recompute of a 60-fund, 15-year, 5,000-transaction portfolio in < 30s; incremental daily NAV update in < 5s.
- **Reliability:** NAV/benchmark fetch failures degrade gracefully (stale-data banner, never silent).
- **Security:** Encrypted local DB; PDF passwords never persisted; LLM calls exclude PII (names, folio numbers, PAN masked).
- **Portability:** Runs on macOS (Mac Mini/MacBook) with no cloud dependency except NAV feeds and optional LLM.

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7. Proposed Architecture (v1)

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CAS PDFs ──► Ingestion (casparser) ──► Raw transaction store (immutable)

                                       │
AMFI NAV feed ──► NAV history store
```



- **Stack suggestion:** Python (casparser, pandas/duckdb, scipy for XIRR) + SQLite; FastAPI + lightweight React front-end; Claude API for the analysis layer; report rendering via HTML→PDF.
- **Why local-first:** aligns with privacy NFRs, zero hosting cost, and your existing casparser +

SQLite pipeline becomes the v1 ingestion module.

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8. Phased Roadmap

Phase	Scope	Outcome
P0 — Truth engine (4–6 wks)	FR-1.1/1.2/1.4, FR-2.1/2.2, FR-3.1/3.2, FR-10.2	Trusted XIRR + history + capital gains from CAS
P1 — Judgment (4 wks)	FR-4 (benchmarks), FR-6.1/6.2 (verdicts + tax), FR-7.1, Direct/Regular audit	"Hold or sell, and what it costs me"
P2 — Intelligence (4 wks)	FR-5 (AI layer), FR-9 (alternatives), FR-8 (goals), FR-10.1 (yearly report)	Full advisor experience

Phase	Scope	Outcome
P3 — Polish	Rolling returns, overlap %, Monte Carlo, digests, watchlist	Depth & retention

MVP definition: end of P1 — because a tool that computes honest XIRR, benchmarks it, and prices the tax of every exit already beats every free tool available.

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9. Success Metrics

- 100% of transactions from CAS reconciled (invested amount matches CAS summary to the rupee).
- XIRR parity with Excel/Zerodha Coin on overlapping funds (±0.05%).
- Time to full portfolio insight from a fresh CAS upload: < 5 minutes.
- ≥ 1 acted-upon recommendation per quarter with documented tax-aware rationale (the real ROI).
- Capital gains report used directly for ITR filing without manual correction.

10. Key Risks & Mitigations

Risk	Mitigation
CAS format changes break parsing	Pin casparser version; golden-file regression tests per CAS vintage
TRI benchmark data hard to source freely	Cache historical TRI; fall back to index-fund NAV as TRI proxy (documented)
LLM hallucinating numbers	Hard boundary (FR-5.2): LLM never computes; validator rejects any number not present in input JSON
Tax rules change (budget cycles)	Assumption registry (§5.7); tax rules as versioned config, not code constants
Scope creep toward robo-advisory SaaS	Non-goals (§2) + compliance gate (§11)

11. Compliance Note

For **personal use**, none of this is regulated. If this is ever offered to others (even free), per-fund buy/sell

recommendations constitute investment advice under SEBI's IA Regulations and require RIA registration; the alternatives/verdict features would need to be reframed as "research/analytics" or gated. Decision point deferred until any multi-user ambition.

12. Resolved Decisions

1. **Asset scope: MF only.** No external assets (EPF/FD/stocks) in v1 — goal planning math uses MF corpus exclusively. FR-1.5's "external corpus" input and FR-8.1's external-corpus field are removed from scope.
2. **Portfolio model: per-PAN.** Each PAN is a separate portfolio with its own tax computation (LTCG exemption, capital gains report per PAN). An optional household roll-up view aggregates metrics (value, XIRR, allocation) across PANs for display only — never for tax math. CAS ingestion tags every folio to a PAN.
3. **Default benchmark: Nifty 50 TRI** for portfolio-level comparison (FR-4.2), user-overridable. Category-level comparisons still use category-correct TRI indices (FR-4.3).

4. **LLM analysis frequency: both.** Auto-run on each monthly CAS refresh (generates the monthly narrative) plus on-demand re-analysis anytime, e.g., after a what-if change or verdict review.